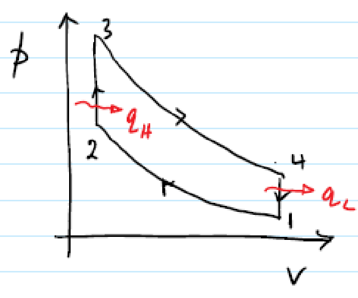
Previous Test and Exam Problems - Solution

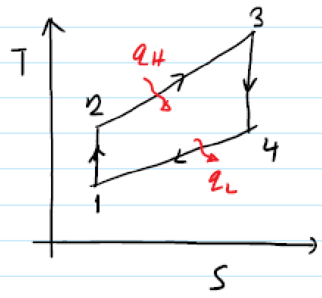
Otto/Diesel Cycle

Problem 1

Otto Cycle

(a)





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1 → 2 isentropic compression
2 → 3 constant volume heat addition
3 → 4 isentropic expansion
4 → 1 constant volume heat rejection

2 (/4)

Given: $r = 8.5 = \frac{V_1}{V_2} = \frac{V_4}{V_3}$ $P_1 = 100\,kPa$, $T_1 = 20°C = 293\,K$

$q_H = 800\,kJ/kg$

Cold-air standard assumptions: $k = 1.4$, $c_p = 1.005\,\frac{kJ}{kg\,K}$ $c_v = 0.718\,\frac{kJ}{kg\,k}$

(b) 1 → 2 isentropic $\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1} \Rightarrow T_2 = 293\,(8.5)^{0.4} = \underline{689.66\,K}$ 1

and $\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{k} \Rightarrow P_2 = 100\,(8.5)^{1.4} = \underline{2000.72\,kPa}$ 1

2 → 3 heat addition: $q_H = c_v(T_3 - T_2)$

$\Rightarrow T_3 = T_2 + \frac{q_H}{c_v} = 689.66 + \frac{800}{0.718} = \underline{\underline{1803.9\,K}}$ 2

const. volume $\Rightarrow \frac{P_3}{T_3} = \frac{P_2}{T_2} \Rightarrow P_3 = P_2 \cdot \frac{T_3}{T_2} = 2000.72 \times \frac{1803.9}{689.66} = \underline{\underline{5233.1\,kPa}}$ 2 (/6)

(c) 3 → 4 isentropic $\Rightarrow \frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{k-1} \Rightarrow T_4 = 1803.9\left(\frac{1}{8.5}\right)^{0.4} = \underline{\underline{766.37\,K}}$ 1

4 → 1 heat rejection: $q_L = c_v(T_4 - T_1) = 0.718(766.37 - 293)$

$\Rightarrow q_L = \underline{\underline{339.88 \ \frac{kJ}{kg}}}$ 2(1/3)

(d) $w_{net,out} = q_H - q_L = 800 - 339.88 = \underline{\underline{460.12 \ \frac{kJ}{kg}}}$ (1/2)

(e) $\eta_{th} = \frac{w_{net\,out}}{q_H} = \frac{460.12}{800} = 0.575 = \underline{\underline{57.5\,\%}}$ 1 (1/2)

check: $\eta_{th,\,otto} = 1 - \frac{1}{r^{k-1}} = 1 - \frac{1}{8.5^{0.4}} = \underline{0.575}$ ✓✓ 1

(f) $MEP = \frac{w_{net,out}}{v_1 - v_2} = \frac{w_{net\,out}}{v_1(1 - \frac{1}{r})}$ and $v_1 = \frac{RT_1}{P_1} = \frac{0.287 \cdot 293}{100}$

$\Rightarrow v_1 = \underline{0.8409} \ m^3/kg$ 1

$\Rightarrow MEP = \frac{460.12}{0.8409(1 - \frac{1}{8.5})} = \underline{\underline{620.1 kPa}}$ 2 (1/3)

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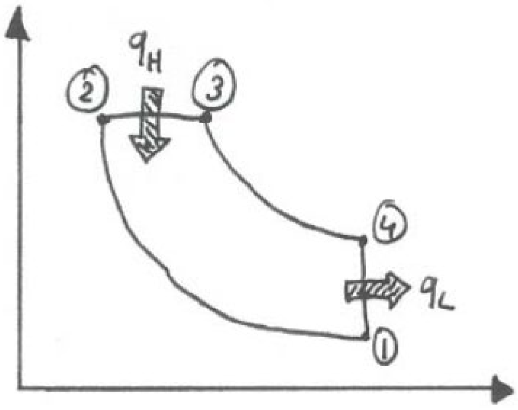
Problem 2



Air standard Diesel cycle

Initial: $P_1 = 98 \ kPa$

$T_1 = 25°C = 298 \ K$

$r = 20 = \frac{V_1}{V_2}$

$r_c = 2.2 = \frac{V_3}{V_2}$

1 → 2 isentropic ⇒

$T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{k-1} = (298 \ K)(20)^{0.4} = 987.71 \ K$ (1) (1)

$$p_2 = p_1\left(\frac{V_1}{V_2}\right)^k = (98\ kPa)(20)^{1.4} = 6496.33\ kPa$$

$2 \to 3$ isobaric $\Rightarrow$

a) $$\frac{p_2 V_2}{T_2} = \frac{p_3 V_3}{T_3} \Rightarrow \underline{T_3} = T_2 \frac{V_3}{V_2} = 987.71 \times 2.2 = \underline{\underline{= 2172.96\ K}}$$

b) $$\underline{q_H} = c_p (T_3 - T_2) = \left(1.005\ \frac{kJ}{kg\,K}\right)(2172.96\ K - 987.71\ K) = \underline{\underline{1191.2\ \frac{kJ}{kg}}}$$

(heat addition)

c) $3 \to 4$ isentropic $\Rightarrow$

$$\frac{V_4}{V_3} = \frac{V_4}{V_2}\cdot\frac{V_2}{V_3} = \frac{V_1}{V_2}\,\frac{V_2}{V_3} = \frac{r}{r_c}$$

$$\underline{\underline{T_4}} = T_3\left(\frac{V_3}{V_4}\right)^{k-1} = (2172.96\ K)\left(\frac{2.2}{20}\right)^{0.4} = \underline{\underline{898.69\ K}}$$

d) $4 \to 1$ isochoric $\Rightarrow$

$$q_L = c_v (T_4 - T_1) = \left(0.718\ \frac{kJ}{kg\,K}\right)(898.69 - 298)\,K = 431.29\ \frac{kJ}{kg}$$

$$\underline{\underline{w_{net}}} = q_H - q_L = (1191.2 - 431.29)\ kJ/kg = \underline{\underline{759.88\ \frac{kJ}{kg}}}$$

e) $$\underline{\underline{\eta_{th}}} = \frac{w_{net}}{q_H} = \frac{759.88}{1191.2} = \underline{\underline{63.8\ \%}}$$

Check: $$\underline{\underline{\eta_{th,Diesel}}} = 1 - \frac{1}{r^{k-1}}\left[\frac{r_c^{\,k} - 1}{k(r_c - 1)}\right] =$$

$$= 1 - \frac{1}{20^{0.4}}\left[\frac{2.2^{1.4}-1}{1.4(2.2-1)}\right] =$$

$$= 0.638 = \underline{\underline{63.8\%}} \quad ✓ \; 2$$

f)

$$MEP \overset{1}{=} \frac{w_{net}}{v_1 - v_2} = \frac{w_{net}}{v_1\left(1-\frac{1}{r}\right)}$$

$$P_1 v_1 = RT_1 \Rightarrow v_1 \overset{1}{=} \frac{RT_1}{P_1} = \frac{\left(0.287\,\frac{kJ}{kg\,K}\right)(298\,K)}{(98\,kPa)} =$$

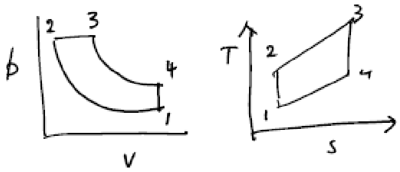
$$= 0.8727\,\frac{m^3}{kg} \quad 1$$

$$\Rightarrow \underline{\underline{MEP}} = \frac{759.88\,\frac{kJ}{kg}}{\left(0.8727\,\frac{m^3}{kg}\right)\left(1-\frac{1}{20}\right)} = \underline{\underline{916.5\,kPa}} \quad 1$$

$\Sigma 20$

Problem 3

Diesel Cycle



$P_1 = 100\,kPa$

$T_1 = 15°C = 288\,K$

$\frac{V_1}{V_2} = r = 19.5\,, \quad T_3 = 1900\,K$

$k = 1.4$
$C_p = 1.005\,kJ/kg\,K$
$C_v = 0.718\,kJ/kg\,K$

(a) $1 \rightarrow 2$ isentropic compression $\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1} \Rightarrow T_2 = 288(19.5)^{0.4}$

$$\Rightarrow T_2 = 944.94\,K \quad 2$$

$2 \rightarrow 3$ constant pressure process $\Rightarrow \frac{\cancel{P_2}V_2}{T_2} = \frac{\cancel{P_3}V_3}{T_3} \Rightarrow \frac{V_3}{V_2} = \frac{T_3}{T_2}$

Now, the cut-off ratio, $r_c = \frac{V_3}{V_2} \Rightarrow r_c = \frac{T_3}{T_2} = \frac{1900}{944.94} = \underline{\underline{2.01}}$ 2

(4 marks)

(b) $q_H = c_p(T_3 - T_2) = 1.005(1900 - 944.94) = \underline{\underline{959.83 \frac{kJ}{kg}}}$ 2

(2 marks)

(c) $3 \rightarrow 4$, isentropic expansion $\Rightarrow \frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{k-1}$

Now $\frac{V_3}{V_4} = \frac{V_3}{V_2} \cdot \frac{V_2}{V_4} = r_c \cdot \frac{1}{r} = \frac{r_c}{r}$

$\Rightarrow T_4 = 1900\left(\frac{2.01}{19.5}\right)^{0.4} = \underline{\underline{765.74\ K}}$ 2 (2 marks)

(d) $4 \rightarrow 1$ const vol, heat rejection

$\Rightarrow q_L = c_v(T_4 - T_1) = 0.718(765.74 - 288) = \underline{343.01 \frac{kJ}{kg}}$ 2

$\Rightarrow w_{net,out} = q_H - q_L$

$= 959.83 - 343.01$

$= \underline{\underline{616.82\ kJ/kg}}$ 2 (4 marks)

(e) $\eta_{th} = \frac{w_{net,out}}{q_H} = \frac{616.82}{959.83} \times 100\% = \underline{\underline{64.3\%}}$ 2

Check: $\eta_{th,\,Diesel} = 1 - \frac{1}{r^{k-1}}\left[\frac{r_c^k - 1}{k(r_c - 1)}\right]$

$= 1 - \frac{1}{19.5^{0.4}}\left[\frac{2.01^{1.4} - 1}{1.4(2.01 - 1)}\right]$

$= 0.6427$ 2 (4 marks)

$= \underline{64.3\%}$ ✓✓

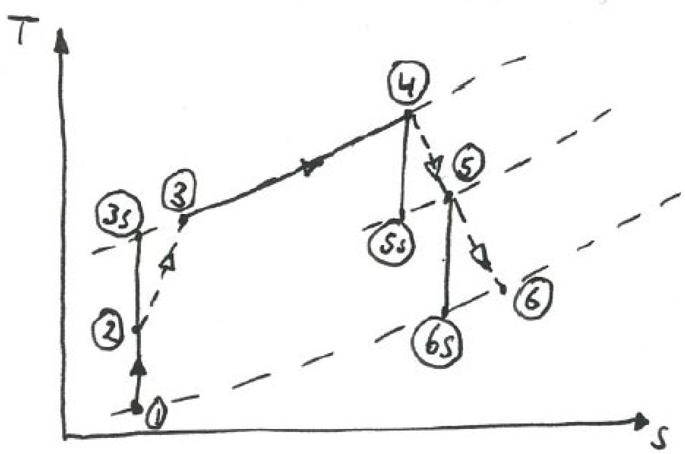
(f) $MEP = \frac{w_{net,out}}{v_{max} - v_{min}} = \frac{w_{net,out}}{v_1 - v_2}$ $\quad v_1 = \frac{RT_1}{P_1} = \frac{0.287 \times 288}{100} = 0.82656 \frac{m^3}{kg}$ 2

$\Rightarrow MEP = \frac{w_{net,out}}{v_1(1 - \frac{1}{r})} = \frac{616.82}{0.82656(1 - \frac{1}{19.5})} = \underline{\underline{786.6\ kPa}}$ 2 (4 marks)

Brayton Cycle

Problem 4

a)



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b)

1 → 2: (diffuser)

$$0 = (h_2 - h_1) + \frac{1}{2}(V_2^2 - V_1^2)$$

$$0 = c_p(T_2 - T_1) + \frac{1}{2}(V_2^2 - V_1^2) \Rightarrow T_2 = T_1 + \frac{1}{2c_p}(V_1^2 - V_2^2)$$ 1

$$T_2 = (273 - 35)\,\text{K} + \frac{1}{2\left(1005\,\frac{\text{J}}{\text{kg K}}\right)}\left[\left(\frac{800\,\frac{\text{km}}{\text{h}} \times 1000\,\frac{\text{m}}{\text{km}}}{3600\,\frac{\text{s}}{\text{h}}}\right)^2 - \left(15\,\frac{\text{m}}{\text{s}}\right)^2\right] =$$

$$= 263\,\text{K}$$ 1

2 → 3: (compressor)

$$T_{3s} = T_2\left(\frac{P_3}{P_2}\right)^{\frac{k-1}{k}} = (263\,\text{K})\left(\frac{450}{50}\right)^{\frac{1.4-1}{1.4}} = 504\,\text{K}$$ 1

$$\eta_c = \frac{h_{3s} - h_2}{h_3 - h_2}$$

$$w_c = h_3 - h_2 = \frac{(h_{3s} - h_2)}{\eta_c} = \frac{c_p(T_{3s} - T_2)}{\eta_c} =$$

$$= \left(1.005\,\frac{\text{kJ}}{\text{kg K}}\right)\frac{(504 - 263)\,\text{K}}{0.83} = 284.6\,\frac{\text{kJ}}{\text{kg}}$$ 1

<u>4 → 5:</u> (turbine)

$$\eta_T = \frac{h_4 - h_5}{h_4 - h_{5s}}$$

$$w_T = h_4 - h_5 = w_c$$ 1

$$\eta_T = \frac{w_c}{c_p (T_4 - T_{5s})} \Rightarrow T_{5s} = T_4 - \frac{w_c}{\eta_T c_p}$$

$$T_{5s} = (950 + 273)\,K - \frac{284.6 \frac{kJ}{kg}}{0.83 (1.005 \frac{kJ}{kgK})} = 881.8\,K$$ 1

$$\underline{\underline{P_5}} = P_4 \left(\frac{T_{5s}}{T_4}\right)^{\frac{k}{k-1}} = (450\,kPa) \left[\frac{881.8}{(950+273)}\right]^{\frac{1.4}{0.4}} = \underline{\underline{142.8\,kPa}}$$ 1

$$\dot{W} = \dot{m}\, w_c \Rightarrow \underline{\underline{\dot{m}}} = \frac{500\,kW}{284.6 \frac{kJ}{kg}} = \underline{\underline{1.76\,\frac{kg}{s}}}$$ 1 1

<u>5 → 6:</u> (nozzle)

$$0 = (h_6 - h_5) + \frac{1}{2}\left(V_6^2 - \underbrace{\cancel{V_5^2}}_{\approx 0}\right)$$ 1

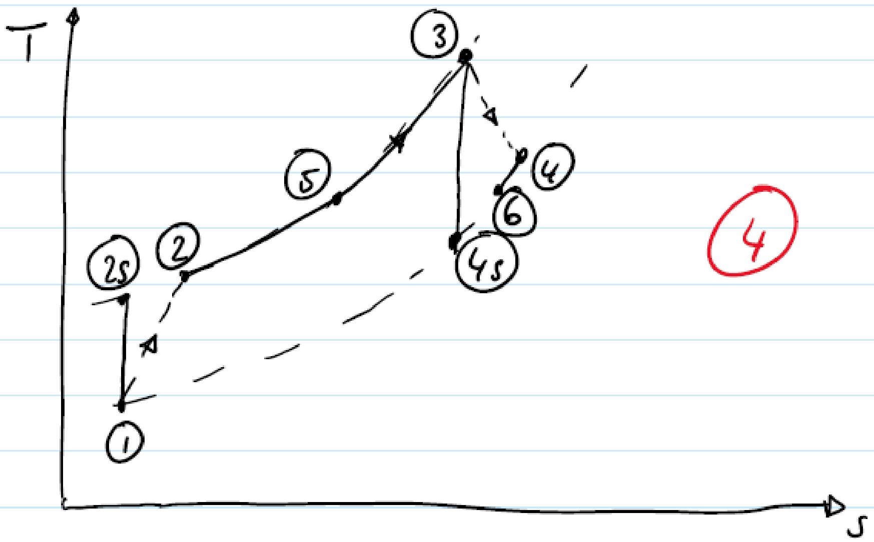
$$V_6 = \sqrt{2 c_p (T_5 - T_6)}$$ 1

$$w_T = w_c = c_p (T_4 - T_5) \Rightarrow T_5 = c_p T_4 - w_c$$

$$T_5 = (1.005 \tfrac{kJ}{kgK})(950 + 273)\,K - 284.6 \tfrac{kJ}{kg} = 944.5\,K$$ 1

Problem 5

a)



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b)

$$T_{2s} = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} = (283\,K)(8)^{\frac{1.4-1}{1.4}} = 512.6\ K$$ 1

$$\eta_c = \frac{h_{2s} - h_1}{h_2 - h_1} = \frac{T_{2s} - T_1}{T_2 - T_1} \Rightarrow T_2 = T_1 + \frac{T_{2s} - T_1}{\eta_c}$$ 1

$$T_2 = (283\,K) + \frac{(512.6 - 283)K}{0.9} = 538.1\ K$$ 1

$$T_{4s} = T_3 \left(\frac{P_1}{P_2}\right)^{\frac{k-1}{k}} = (1088\,K)\left(\frac{1}{8}\right)^{\frac{1.4-1}{1.4}} = 600.6\ K$$ 1 1

$$\eta_T = \frac{h_3 - h_4}{h_3 - h_{4s}} = \frac{T_3 - T_4}{T_3 - T_{4s}} \Rightarrow T_4 = T_3 - \eta_T (T_3 - T_{4s})$$ 1

$$T_4 = (1088\,K) - 0.9(1088 - 600.6)K = 649.34\ K$$ 1

$$w_{net} = (h_3 - h_4) - (h_2 - h_1) = c_p (T_3 - T_4 - T_2 + T_1) =$$ 1

$$= \left(1.005 \frac{kJ}{kg\,K}\right)(1088 - 649.34 - 538.1 + 283)K = 184.5 \frac{kJ}{kg}$$ 1

c) Regenerator effectiveness $\varepsilon = \frac{h_5 - h_2}{h_4 - h_2} = \frac{T_5 - T_2}{T_4 - T_2}$ 1

$T_5 = T_2 + \varepsilon (T_4 - T_2) = (538.1\,K) + 0.85(649.3 - 538.1)\,K =$

$= 632.6\,K$ 1

$\eta_{th,reg} = \frac{w_{net}}{q_{in}} = \frac{w_{net}}{c_p(T_3 - T_5)} = \frac{184.5\,\frac{kJ}{kg}}{(1.005\,\frac{kJ}{kg\,K})(1088 - 632.6)\,K} =$ 1

$= 0.403 = \underline{40.3\,\%}$ 1

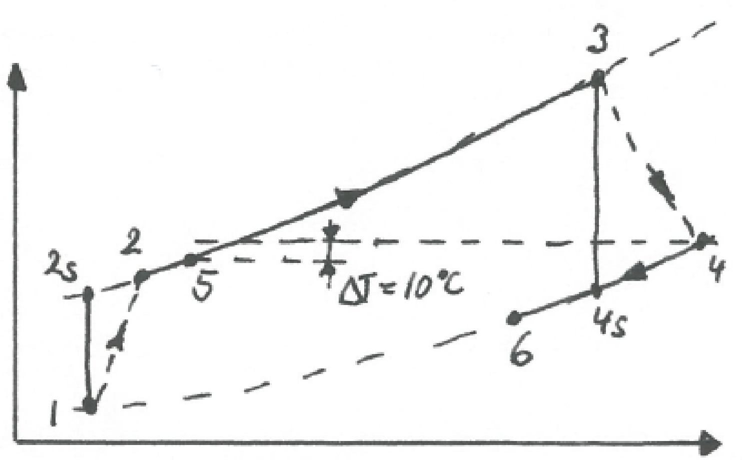
d) Without regenerator:

$\eta_{th,no\,reg} = \frac{w_{net}}{q_{in}} = \frac{w_{net}}{c_p(T_3 - T_2)} = \frac{184.5\,\frac{kJ}{kg}}{(1.005\,\frac{kJ}{kg\,K})(1088 - 538.1)\,K} =$ 1

$= 0.334 = \underline{33.4\,\%}$ 1

Problem 6

a)



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b) $T_{2s} = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{k-1}{k}} = (293\ K)(8)^{\frac{0.4}{1.4}} = 530.8\ K$ (1)

$\eta_c = \frac{c_p(T_{2s}-T_1)}{c_p(T_2-T_1)} \Rightarrow T_2 = (293\ K) + \frac{(530.8-293)K}{0.87} = 566.3\ K$ (1)

$T_{4s} = T_3 \left(\frac{P_1}{P_2}\right)^{\frac{k-1}{k}} = (1073\ K)\left(\frac{1}{8}\right)^{\frac{0.4}{1.4}} = 592.3\ K$ (1)

$\eta_T = \frac{c_p(T_3-T_4)}{c_p(T_3-T_{4s})} \Rightarrow T_4 = (1073\ K) - 0.93(1073-592.3)K = 625.9\ K$ (1)

$w_{net} \overset{(1)}{=} c_p(T_3-T_4) - c_p(T_2-T_1) = \left(1.005\ \frac{kJ}{kgK}\right)(1073-625.9-566.3+293)K =$

$= 174.7\ kJ/kg$ (1)

$\underline{\underline{\dot{m}}} \overset{(1)}{=} \frac{\dot{W}_{net}}{w_{net}} = \frac{150\ kW}{174.7\ kJ/kg} = \underline{\underline{0.859\ kg/s}}$ (1)

c) $T_5 = T_4 - \Delta T = (625.9-10)K = 615.9\ K$

$q_{in} = c_p(T_3 - T_5) = \left(1.005\ \frac{kJ}{kgK}\right)(1073-615.9)K = 459.4\ \frac{kJ}{kg}$ (1)

$\underline{\underline{\eta_{th}}} \overset{(1)}{=} \frac{w_{net}}{q_{in}} = \frac{174.7}{459.4} = 0.38 = \underline{\underline{38\ \%}}$ (1)

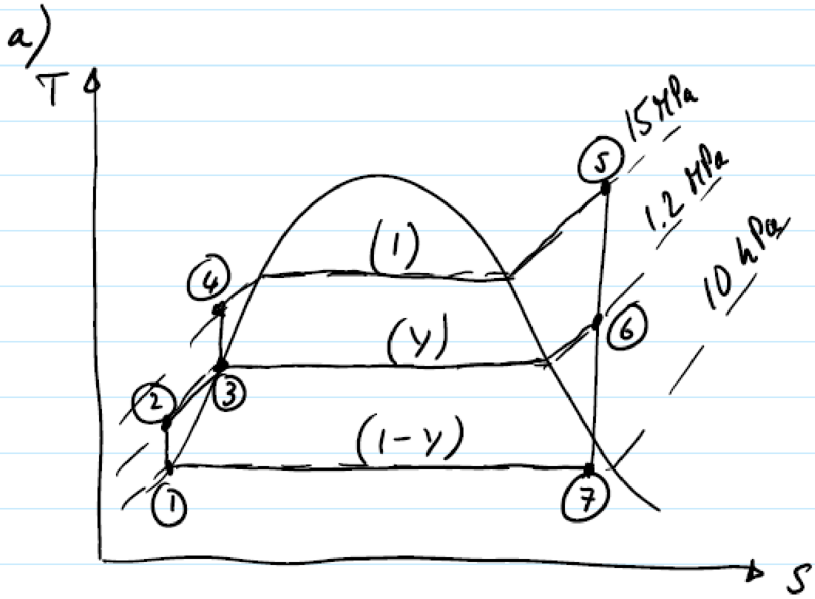
d) $\underline{\underline{\varepsilon}} \overset{(1)}{=} \frac{T_5-T_2}{T_4-T_2} = \frac{(615.9-566.3)}{(625.9-566.3)} = 0.83 = \underline{\underline{83\ \%}}$ (1)

$\Sigma 15$

Rankine Cycle

Problem 7

a)



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b)

$$h_1 = h_{f@10\,kPa} = 191.81 \ \frac{kJ}{kg} \quad 1$$

$$v_1 = v_{f@10\,kPa} = 0.001010 \ \frac{m^3}{kg} \quad 1$$

$$h_2 \overset{1}{=} h_1 + v_1 (P_2 - P_1) = 191.81 \ \frac{kJ}{kg} + 0.001010 \ \frac{m^3}{kg} (1200 - 10)\ kPa =$$

$$= 193.01 \ \frac{kJ}{kg} \quad 1$$

$$h_3 = h_{f@1.2\,MPa} = 798.33 \ \frac{kJ}{kg} \quad 1$$

$$h_3 = h_{f@1.2\,MPa} = 798.33 \ \frac{kJ}{kg} \quad 1$$

$$v_3 = v_{f@1.2\,MPa} = 0.001138 \ \frac{m^3}{kg} \quad 1$$

$$h_4 \overset{1}{=} h_3 + v_3 (P_4 - P_3) = 798.33 \ \frac{kJ}{kg} + 0.001138 \ \frac{m^3}{kg} (15000 - 1200)\ kPa =$$

$$= 814.03 \ \frac{kJ}{kg} \quad 1$$

Open FWH: $y h_6 + (1-y) h_2 = h_3$ (1)

$$y h_6 + h_2 - y h_2 = h_3$$

$$y = \frac{h_3 - h_2}{h_6 - h_2}$$

Finding h_6:

$$\left.\begin{aligned} P_5 &= 15\ \text{MPa} \\ T_5 &= 600^\circ\text{C} \end{aligned}\right\} \quad \begin{aligned} h_5 &= 3583.1\ \tfrac{\text{kJ}}{\text{kg}} \ (1) \\ s_5 &= 6.6796\ \tfrac{\text{kJ}}{\text{kg K}} \ (1) \end{aligned}$$

$$\left.\begin{aligned} P_6 &= 1.2\ \text{MPa} \\ s_6 &= s_5 \ (1) \end{aligned}\right\} \quad \begin{aligned} h_6 &= 2860.2\ \tfrac{\text{kJ}}{\text{kg}} \ (1) \\ &(\text{by interpolation}) \end{aligned}$$

$$\underline{\underline{y}} = \frac{798.33\ \frac{\text{kJ}}{\text{kg}} - 193.01\ \frac{\text{kJ}}{\text{kg}}}{2860.1\ \frac{\text{kJ}}{\text{kg}} - 193.01\ \frac{\text{kJ}}{\text{kg}}} = 0.227 = \underline{\underline{22.7\ \%}} \ (1)$$

c) $\dot{W}_{net} = \dot{m}\left[(h_5 - h_6) + (1-y)(h_6 - h_7) - (h_4 - h_3) - (1-y)(h_2 - h_1)\right]$ (1)

Finding h_7:

$s_7 = s_5$ (1) $= 6.6796\ \frac{\text{kJ}}{\text{kg K}}$; $s_7 = s_f + x_7\, s_{fg}$ (1)

$$\Rightarrow x_7 \ (1) = \frac{s_7 - s_f}{s_{fg}} = \frac{6.6796 - 0.6492}{7.4996\ (1)} = 0.80 \ (1)$$

h_7 (1) $= h_f + x_7 h_{fg} = 191.81\ \frac{\text{kJ}}{\text{kg}} + 0.80\left(2392.1\ \frac{\text{kJ}}{\text{kg}}\right) = 2105.49\ \frac{\text{kJ}}{\text{kg}}$ (1)

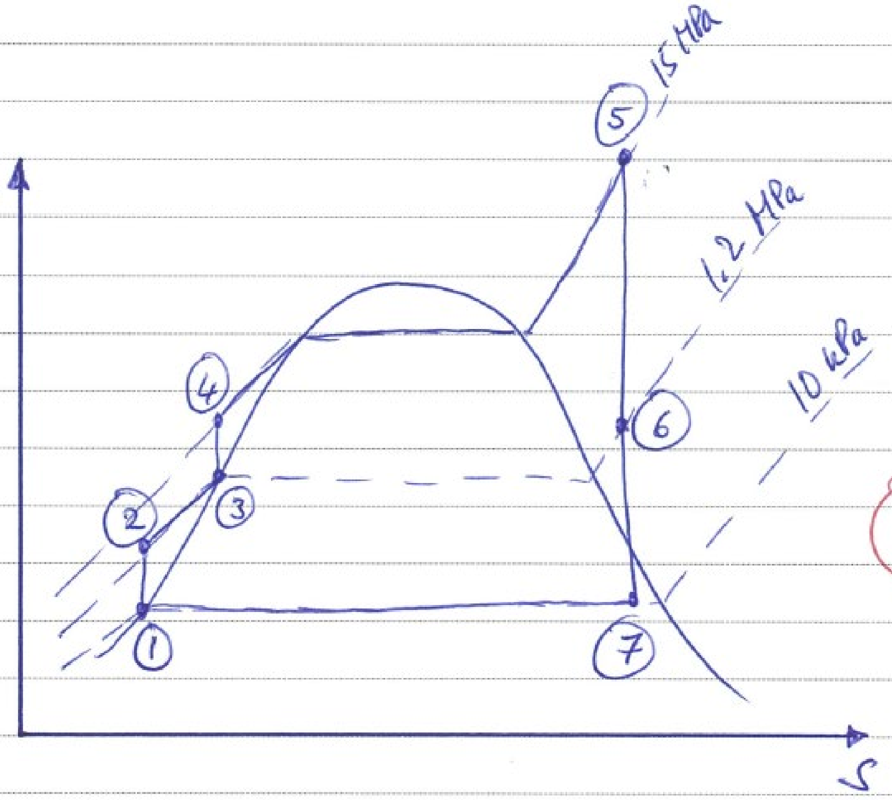
$$\underline{\underline{\dot{W}_{net}}} = 84\ \frac{\text{kg}}{\text{s}}\Big[(3583.1 - 2860.2) + (1 - 0.227)(2860.2 - 2105.49)$$
$$- (814.03 - 798.33) - (1 - 0.227)(193.01 - 191.81)\Big]\ \frac{\text{kJ}}{\text{kg}} =$$

$$= \underline{\underline{108.3\ \text{MW}}} \ (1)$$

$\Sigma 27$

Problem 8

a)



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b)

$h_1 = h_{f @ 10\,kPa} = 191.81\ \frac{kJ}{kg}$ 1

$v_1 = v_{f @ 10\,kPa} = 0.001010\ \frac{m^3}{kg}$ 1

$h_2 = h_1 + v_1 (P_2 - P_1) =$ 1

$= \left(191.81\ \frac{kJ}{kg}\right) + \left(0.001010\ \frac{m^3}{kg}\right)(1200 - 10)\,kPa =$

$= 193.01\ \frac{kJ}{kg}$ 1

$h_3 = h_{f @ 1.2\,MPa} = 798.33\ \frac{kJ}{kg}$ 1

$v_3 = v_{f @ 1.2\,MPa} = 0.001138\ \frac{m^3}{kg}$ 1

$h_4 = h_3 + v_3 (P_4 - P_3) =$ 1

$$= \left(798.33\ \frac{kJ}{kg}\right) + \left(0.001138\ \frac{m^3}{kg}\right)(15000 - 1200)\ kPa =$$

$$= 814.03\ \frac{kJ}{kg} \quad 1$$

Open FWH: $\quad y h_6 + (1-y) h_2 = h_3 \quad 1$

$$y h_6 + h_2 - y h_2 = h_3$$

$$y = \frac{h_3 - h_2}{h_6 - h_2} \quad 1$$

Finding h_6:

$$\left.\begin{array}{l} P_5 = 15\ MPa \\ T_5 = 600°C \end{array}\right\} \begin{array}{l} h_5 = 3583.1\ \frac{kJ}{kg} \quad 1 \\ s_5 = 6.6796\ \frac{kJ}{kg} \quad 1 \end{array}$$

$$\left.\begin{array}{l} P_6 = 1.2\ MPa \\ s_6 = s_5 \end{array}\right\} \begin{array}{l} s_5 > s_g @ 1.2 MPa \Rightarrow \text{superheated} \\ h_6 \overset{1}{=} 2860.2\ \frac{kJ}{kg}\ (\text{by interpolation}) \end{array}$$

$$\underline{\underline{y}} = \frac{\left(798.33\ \frac{kJ}{kg}\right) - \left(193.01\ \frac{kJ}{kg}\right)}{\left(2860.1\ \frac{kJ}{kg}\right) - \left(193.01\ \frac{kJ}{kg}\right)} = \underline{\underline{0.227}} \quad 1$$

c)

$$\dot{W}_{net} = \dot{m}\left[(h_5 - h_6) + (1-y)(h_6 - h_7) - (h_4 - h_3) - (1-y)(h_2 - h_1)\right] \quad 1$$

Finding h_7:

$$s_7 = s_5 = 6.6796 \ \frac{kJ}{kg\,K} \quad 1$$

$$s_7 = s_f + x_7 s_{fg} \Rightarrow x_7 = \frac{s_7 - s_g}{s_{fg}} = \quad 1$$

$$= \frac{6.6796 - 0.6492}{7.4996} = 0.80 \quad 1$$

$$h_7 = h_f + x_7 h_{fg} = \quad 1$$

$$= \left(191.81 \frac{kJ}{kg}\right) + 0.80\left(2392.1 \frac{kJ}{kg}\right) = 2105.49 \ \frac{kJ}{kg} \quad 1$$

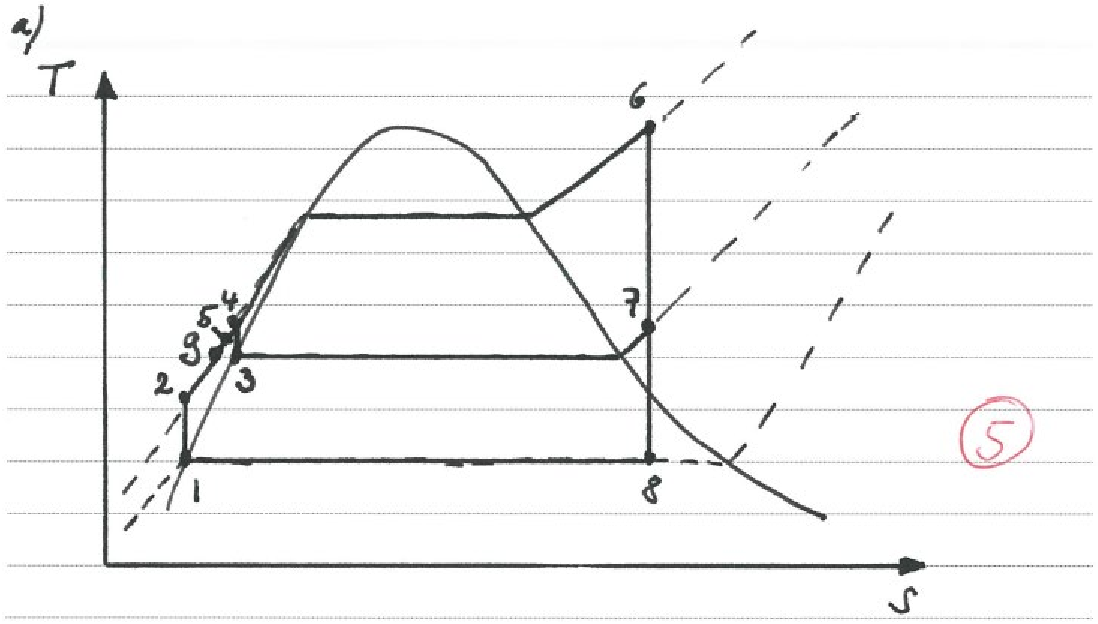
$$\underline{\underline{\dot{W}_{net}}} = \left(84 \frac{kg}{s}\right)\Big[(3583.1 - 2860.2) + (1 - 0.227)(2860.2 - 2105.49)$$
$$- (814.03 - 798.33) - (1 - 0.227)(193.01 - 191.81)\Big] \frac{kJ}{kg} =$$

$$= \underline{\underline{108.3 \ MW}} \quad 1$$

$\Sigma 25$

Problem 9

a)



b)

$$h_1 = h_{f@20\,kPa} = 251.42\ \frac{kJ}{kg} \quad (1)$$

$$v_1 = v_{f@20\,kPa} = 0.001017\ m^3/kg \quad (1)$$

$$w_{p1} \overset{(1)}{=} v_1(P_2 - P_1) = \left(0.001017\ \frac{m^3}{kg}\right)(8000-20)\,kPa = 8.12\ \frac{kJ}{kg} \quad (1)$$

$$h_2 \overset{(1)}{=} h_1 + w_{p1} = (251.42 + 8.12)\ \frac{kJ}{kg} = 259.54\ \frac{kJ}{kg} \quad (1)$$

$$\left.\begin{array}{l} P_6 = 8\,MPa \\ T_6 = 450°C \end{array}\right\} \quad \begin{array}{l} h_6 = 3273.3\ \frac{kJ}{kg} \quad (1) \\ s_6 = 6.5579\ \frac{kJ}{kg\,K} \quad (1) \end{array}$$

$$\left.\begin{array}{l} P_7 = 2\,MPa \\ s_7 = s_6 \end{array}\right\} \quad h_7 = 2903.3 + (6.5579 - 6.5475) \times \frac{(3024.2 - 2903.3)}{(6.7684 - 6.5475)} =$$

$$= 2908.99\ \frac{kJ}{kg} \quad (1)$$

$$\left.\begin{array}{l} P_8 = 20\,kPa \\ s_8 = s_6 \end{array}\right\} \quad x_8 = \frac{s_8 - s_f}{s_{fg}} = \frac{(6.5579 - 0.8320)}{7.0752} =$$

$$= 0.809 \quad (1)$$

$$h_8 = h_f + x_8 h_{fg} = 251.42 + 0.809(2357.5) =$$
$$= 2158.64 \ \frac{kJ}{kg} \quad 1$$

$$\left.\begin{array}{l} P_3 = 2MPa \\ x_3 = 0 \end{array}\right\} \quad \begin{array}{l} h_3 = h_{f@2MPa} = 908.47 \ \frac{kJ}{kg} \quad 1 \\ v_3 = v_{f@2MPa} = 0.001177 \ \frac{m^3}{kg} \quad 1 \end{array}$$

$$w_{p2} \overset{1}{=} v_3(P_4 - P_3) = \left(0.001177 \ \frac{m^3}{kg}\right)(8000 - 2000)\,kPa =$$
$$= 7.06 \ \frac{kJ}{kg} \quad 1$$

$$h_4 \overset{1}{=} h_3 + w_{p2} = (908.47 + 7.06) \ \frac{kJ}{kg} = 915.53 \ \frac{kJ}{kg} \quad 1$$

Mixing chamber: $\Sigma(\dot{m}h)_{in} = \Sigma(\dot{m}h)_{out}$

$$h_5 = (1-y)h_9 + y h_4 \quad 1$$

$$\left.\begin{array}{l} P_9 = 8MPa \\ T_9 = T_{s@2MPa} \\ \quad = 212.38\,°C \end{array}\right\} \quad h_9 = h_{f@2MPa} = 908.47 \ \frac{kJ}{kg} \quad 1$$

$$h_5 = (1-0.25)\left(908.47 \ \frac{kJ}{kg}\right) + 0.25(915.53) \ \frac{kJ}{kg} = 910.24 \ \frac{kJ}{kg} \quad 1$$

$$w_{net} = w_{turb} - w_{p1} - w_{p2} = (h_6 - h_7) + (1-y)(h_7 - h_8) - w_{p1} - w_{p2} =$$
$$= (3273.2 - 2903.3) + 0.75(2903.3 - 2158.64) - 8.12 - 7.06 =$$
$$= 913.22 \ \frac{kJ}{kg} \quad 1$$

$$q_{in} = h_6 - h_5 = (3273.3 - 910.24) \ \frac{kJ}{kg} = 2363.06 \ \frac{kJ}{kg} \quad 1$$

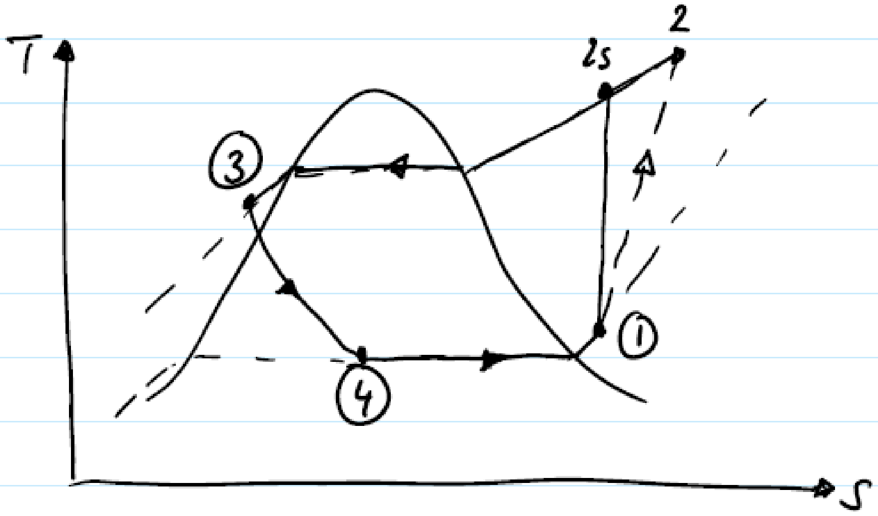
$$\underline{\underline{\eta_{th}}} = \frac{w_{net}}{q_{in}} = \frac{913.22}{2363.06} = 0.387 = \underline{\underline{38.7\ \%}} \quad 1$$

$\Sigma 28$

Refrigeration

Problem 10

a)



$$\left.\begin{array}{l} P_1 = 60 \text{ kPa} \\ T_1 = -20\,°C \end{array}\right\} h_1 = 240.78 \ \frac{kJ}{kg} \quad 1$$

$$\left.\begin{array}{l} P_2 = 1200 \text{ kPa} \\ T_2 = 70\,°C \end{array}\right\} h_2 = 300.63 \ \frac{kJ}{kg} \quad 1$$

$$\left.\begin{array}{l} P_3 = 1200 \text{ kPa} \\ T_3 = 42\,°C \end{array}\right\} h_3 = h_{f@42°C} = 111.28 \ \frac{kJ}{kg} \quad 1$$

$$h_4 = h_3 = 111.28 \ \frac{kJ}{kg} \quad 1$$

$$h_4 = h_f + x_4 h_{fg} \quad 1$$

$$\Rightarrow \underline{\underline{x_4}} = \frac{h_4 - h_f}{h_{fg}} = \frac{111.28 - 3.837}{223.96} = \underline{\underline{0.48}} \quad 1$$

b)

Condenser : $\dot{m}_w (h_{w2} - h_{w1}) = \dot{m}_R (h_2 - h_3)$

$h_{w2} = h_{f@25°C} = 104.83 \frac{kJ}{kg}$; $h_{w1} = h_{f@20°C} = 83.915 \frac{kJ}{kg}$

$$\dot{m}_R = \dot{m}_w \frac{(h_{w2} - h_{w1})}{(h_2 - h_3)} = (0.25 \tfrac{kg}{s}) \frac{(104.83 - 83.915)\frac{kJ}{kg}}{(300.63 - 111.28)\frac{kJ}{kg}} =$$

$$= 0.028 \frac{kg}{s}$$

$$\dot{Q}_H = \dot{m}_R (h_2 - h_3) = (0.028 \tfrac{kg}{s})(300.63 - 111.28)\tfrac{kJ}{kg} = 5.3 \text{ kW}$$

$$\dot{W}_{in} = \dot{m}_R (h_2 - h_1) - \dot{Q}_{in} = (0.028 \tfrac{kg}{s})(300.63 - 240.78)\tfrac{kJ}{kg} - 0.45 \text{ kW} =$$

$$= 1.23 \text{ kW}$$

$$\underline{\underline{\dot{Q}_L}} = \dot{Q}_H - \dot{W}_{in} - \dot{Q}_{in} = (5.3 - 1.23 - 0.45) \text{ kW} = \underline{\underline{3.62 \text{ kW}}}$$

c)

$$\underline{\underline{COP_R}} = \frac{\dot{Q}_L}{\dot{W}_{in}} = \frac{3.62}{1.23} = \underline{\underline{2.94}}$$

Σ 18

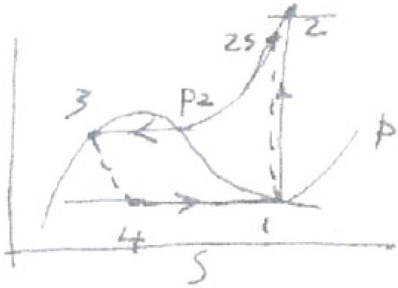
Problem 11

Part A:

- Must evaporate at a pressure about 5 to 10°C below that of fridge space.
- Pressure must not be below atmospheric ∴ will cause air leakage into system.

Part B:

(a) $p_2 = 800\,\text{kPa}$
$p_1 = 140\,\text{kPa}$
$\eta_c = 85\%$
$\dot{Q}_{in} = 250\,\text{kW}$.



2

(b.) From Tables:

$p_1 = 140\,\text{kPa} \} \; h_f = 25.77;\; h_{fg} = 210.27;\; h_g = 236.04 = h_1$
$s_f = 0.1055;\; s_g = 0.9322$

$p_3 = 800\,\text{kPa} \} \; h_f = 93.42$

throttle: $h_3 = h_4 = h_f = 93.42\,\text{kJ/kg}$.

$$x_4 = \frac{h_4 - h_f}{h_{fg}} = \frac{93.42 - 25.77}{210.27} = 0.322$$

5

(c.) $s_1 = s_{2s} = 0.9322$, $p_2 = 800\,\text{kPa}$

s	h
0.9066	264.15
0.9374	273.66

Interpolate: $h_{2s} = 264.15 + (273.7 - 264.2)\left(\dfrac{0.9322 - 0.9066}{0.9374 - 0.9066}\right)$

$= 272.1$ $\qquad h_2 = 278.5$

$\dot{Q}_{in} = 250\,\text{kW} = \dot{m}(h_1 - h_4) \quad \therefore \dot{m} = \dfrac{250 \times 10^3}{(236.04 - 93.42) \times 10^3}$

$= 1.753\,\text{kg/s}$

$$\dot{W}_c = \frac{\dot{m}}{\eta}(h_{2s} - h_1) = \frac{1.753}{0.85}(272.1 - 236.06)$$

$= \underline{74.33\,\text{kW}}$

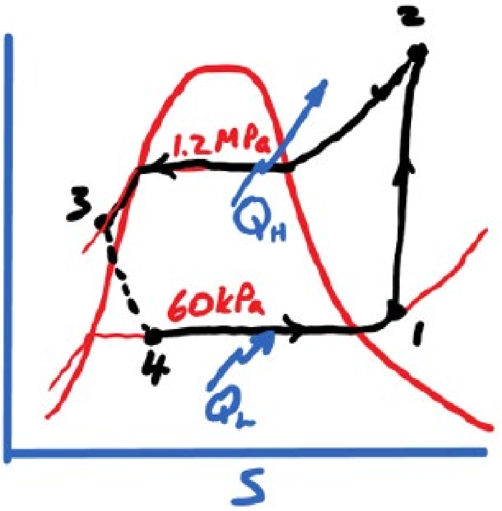
9

(d.) $$COP = \frac{\dot{Q}_{in}}{\dot{W}_c} = \frac{250}{74.33} = \underline{3.36}$$

2

Problem 12

(a) T-s Diagram: R-134a



(b) $\left.\begin{array}{l} p_1 = 60\text{kPa} \\ T_1 = -34°C \end{array}\right\}$ $h_1 = 230.04$

$\left.\begin{array}{l} p_2 = 1.2\text{MPa} \\ T_2 = 65°C \end{array}\right\}$ $h_2 = 295.18$

$\left.\begin{array}{l} p_3 = 1.2\text{MPa} \\ T_3 = 42°C \end{array}\right\}$ $h_3 = h_{f@42°C} = 111.25$

For Water : Condenser

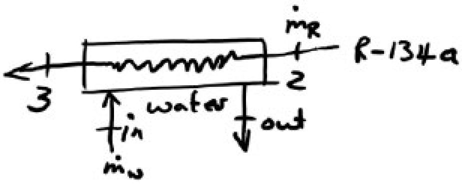
$T_{in} = 18°C \therefore h_{in} = h_{f@18°C} = 75.47$

$T_{out} = 26°C \therefore h_{out} = h_{f@26°C} = 108.94$

$\Delta h = 33.5$ kJ/kg.

very slight difference

Can also use: $\Delta h = C_p \Delta T$
where $C_p = 4.2$ kJ/kgK from table A-3
and $\Delta h = 4.2(26-18) = 33.6$ kJ/kg

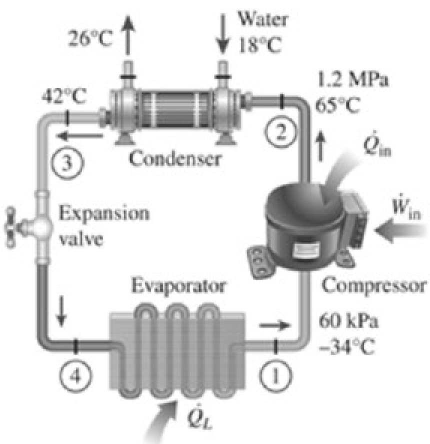


Condenser Energy Balance

$\dot{m}_R(h_2 - h_3) = \dot{m}_w(h_{out} - h_{in})$

$\therefore \dot{m}_R(295.18 - 111.25) = 0.25(108.94 - 75.47)$

$\therefore \dot{m}_R = 0.04549$ kg/s



Cooling load:

$\dot{Q}_L = \dot{m}_R(h_1 - h_4) = 0.0455(230.04 - 111.25)$

$= 5.404$ kW

Compressor Power:

$\dot{Q}_{in} - \dot{W}_{in} = \dot{m}_R(h_2 - h_1)$

$\therefore \dot{W}_{in} = 450 - 0.0455(295.18 - 230.04)$

$= -2.513$ kW

$COP = \left|\frac{\dot{Q}_L}{\dot{W}_{in}}\right| = \frac{5.404}{2.513} = 2.15$

(c.) - Compressors are piston-cylinder devices and cannot deal with any liquid
- Compressors will be badly damaged by liquid at inlet
- To prevent liquid, must superheat
- Ensures only dry-gas in compressor.